

HAMILTON ISLAND, Australia

WMO: 943680

Lat: 20.3662S

Lon: 148.9535E

Elev: 59

StdP: 100.61

Time Zone: 10.00 (E10)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	17.0	4.8	5.4	18.6	7.0	6.3	18.7	16.1	19.0	14.7	19.1	7.7	150	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	30.8	25.6	30.1	25.2	29.4	24.7	26.7	29.0	26.2	28.4	25.9	28.1	5.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	28.1	25.7	21.1	27.8	25.1	20.4	27.5	84.7	29.3	82.3	28.6	80.3	28.3	36.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	14.2	13.2	DB	15.2	33.3	1.2	2.3	14.4	35.0	13.6	36.4	13.0	37.7	12.1	39.4	(4)
(5)			WB	9.2	28.6	1.3	2.9	8.3	30.7	7.6	32.4	6.8	34.0	5.9	36.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	27.2	27.1	26.3	24.8	22.5	20.4	19.7	20.4	22.5	24.3	25.7	27.0		
	DBStd	2.97	1.13	1.16	1.26	1.11	1.40	1.51	1.39	1.39	1.32	1.14	1.11	1.12		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	8	0	0	0	0	0	2	4	2	0	0	0	0		
	CDD10.0	5101	533	479	506	443	387	313	299	323	374	445	470	527		
	CDD18.3	2067	275	246	248	193	129	65	46	66	124	187	220	269		
	CDH23.3	13859	2778	2491	2175	1054	204	25	6	32	238	801	1511	2544		
	CDH26.7	2301	619	549	319	41	1	0	0	0	6	55	177	533		
(14) Wind	WSAvg	7.6	7.8	7.9	8.7	8.5	8.2	7.8	7.4	6.5	6.4	7.3	7.1	7.3		
(15) Precipitation	PrecAvg	1737	293	385	262	174	110	74	42	39	32	39	89	198		
	PrecMax	3241	672	948	996	401	364	237	184	171	264	174	731	1331		
	PrecMin	826	28	62	44	41	18	22	0	0	0	0	0	22		
	PrecStd	596	186	253	227	91	69	52	42	42	51	44	142	246		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	31.2	30.9	28.7	26.8	25.2	24.2	25.4	27.2	29.1	30.2	31.8		
		MCWB	26.3	26.0	26.6	23.9	22.6	21.5	20.2	20.3	21.5	22.9	23.5	25.4		
	2%	DB	30.3	30.2	29.7	27.8	25.8	24.0	23.0	24.1	26.2	28.0	29.1	30.2		
		MCWB	25.7	25.6	25.3	23.5	21.8	21.2	20.1	19.6	21.1	22.6	23.5	24.8		
	5%	DB	29.9	29.9	29.0	27.0	25.0	23.1	22.1	23.2	25.4	27.1	28.2	29.8		
		MCWB	25.3	25.4	24.8	22.9	21.5	20.8	19.4	19.2	20.7	22.3	22.9	24.5		
	10%	DB	29.1	29.1	28.2	26.2	24.2	22.4	21.5	22.3	24.8	26.4	27.8	29.0		
		MCWB	24.8	25.0	24.4	22.7	21.1	20.1	19.0	18.9	20.4	21.6	22.6	23.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.7	27.3	25.7	23.9	23.2	21.7	21.6	23.1	24.5	25.5	26.5		
		MCDB	29.7	29.0	29.0	27.3	25.4	23.9	23.0	24.0	25.6	27.4	28.2	29.6		
	2%	WB	26.7	26.7	26.2	24.7	23.0	22.1	21.0	20.8	22.4	23.7	24.6	25.9		
		MCDB	29.1	28.8	28.0	26.0	24.7	23.2	22.1	22.8	24.9	26.3	27.6	28.7		
	5%	WB	26.2	26.2	25.8	24.1	22.4	21.3	20.5	20.3	21.9	23.2	24.0	25.4		
		MCDB	28.5	28.4	27.7	25.8	24.1	22.6	21.6	22.3	24.3	26.0	27.2	28.2		
	10%	WB	25.7	25.8	25.4	23.6	21.9	20.7	19.8	19.7	21.2	22.5	23.4	24.9		
		MCDB	28.1	28.1	27.3	25.5	23.7	22.1	21.0	21.7	23.7	25.5	26.7	27.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	3.8	3.4	3.1	3.0	2.8	2.9	3.7	4.3	4.4	4.3	4.4		
		MCDBR	4.7	4.4	3.9	3.6	3.4	3.5	3.3	4.5	5.1	5.0	4.8	5.1		
	5% WB	MCWBR	2.4	2.6	2.4	2.4	2.4	2.5	2.4	2.7	2.7	2.4	2.4	2.4		
		MCWBR	4.3	4.1	3.8	3.1	3.1	3.1	3.1	4.1	4.6	4.7	4.8	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.399	0.397	0.388	0.361	0.345	0.333	0.320	0.327	0.355	0.380	0.385	0.395		
		taud	2.542	2.549	2.560	2.620	2.624	2.648	2.663	2.633	2.536	2.492	2.518	2.522		
	Ebn at Noon		946	936	921	907	886	880	904	930	937	941	953	950		
		Edh at Noon	111	109	103	91	84	79	79	88	104	114	113	113		
(44) All-Sky Solar Radiation	RadAvg		6.40	6.05	5.56	4.96	4.28	3.83	4.32	5.15	6.24	6.90	7.04	7.02		
	RadStd		0.53	0.67	0.55	0.45	0.28	0.38	0.31	0.34	0.38	0.38	0.81	0.73		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page